

SIMATS ENGINEERING — Assessment Tool 3

Database Management Systems (CSA05) — CO3: Normalization & Functional Dependencies

MCQ Answer Key with Explanations and SQL Demonstrations

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Each question below shows the correct option, a plain-language explanation — with additional justification points for deeper understanding — and a short SQL demonstration (styled as a terminal/CLI screenshot) illustrating the concept using standard SQL syntax compatible with MySQL. Demonstrations were executed in a SQL engine in this sandbox since a live MySQL server isn't installable here (no network access), but the SQL shown is standard DDL/DML that runs the same way in MySQL.

Q1. Which of the following is true about 1NF?

Bloom's Level: BL2 – Understand | Marks: 5

- (a) Allows multi-valued attributes
- (b) Eliminates partial dependencies
- ✓ **(c) All attributes must be atomic — CORRECT ANSWER**
- (d) Eliminates transitive dependencies

Explanation:

First Normal Form (1NF) requires every column to hold a single, indivisible (atomic) value — no repeating groups or comma-separated lists in one cell. Eliminating partial dependencies is the job of 2NF, and eliminating transitive dependencies is the job of 3NF, so those describe higher normal forms, not 1NF itself.

Additional Justification:

- "Atomic" is relative to the application, not an absolute limit — a value only needs to be split into its own column if the schema will ever need to query or update that part on its own.
- A common mix-up is treating "atomic" as "short." A long text field (e.g., a description) can still be atomic — atomicity is about not packing multiple independent facts into one cell, not about length.
- Exam tip: for a 1NF question, check only for repeating groups or multi-valued cells — ignore any option mentioning keys, partial dependency, or transitive dependency, since those always belong to 2NF/3NF.

Q1 Demo: 1NF requires atomic (single-valued) attributes

```
mysql> SELECT * FROM Student_BadDesign;
+-----+-----+-----+
| StudentID | Name | Phone |
+-----+-----+-----+
| 1         | Arun | 9876543210,9123456780 |
+-----+-----+-----+
1 row in set
```

```
mysql> SELECT * FROM Student_1NF;
```

```
+-----+-----+-----+
| StudentID | Name | Phone |
+-----+-----+-----+
| 1         | Arun | 9876543210 |
| 1         | Arun | 9123456780 |
+-----+-----+-----+
2 rows in set
```

Q2. A relation is in 2NF if it is in 1NF and:

Bloom's Level: BL2 – Understand | Marks: 5

- (a) Has no transitive dependencies
- ✓ **(b) Has no partial dependencies on primary key — CORRECT ANSWER**
- (c) Every attribute is a prime attribute
- (d) Has no multi-valued dependencies

Explanation:

2NF builds on 1NF by additionally requiring that every non-key attribute depend on the WHOLE primary key, not just part of it. A 'partial dependency' happens only when the key is composite (more than one column) and some attribute depends on just one part of that key — as with StudentName depending only on StudentID below, not on the full (StudentID, CourseID) key.

Additional Justification:

- If the primary key is a single column (not composite), the relation automatically satisfies 2NF once it's in 1NF — there's no "part" of a single-column key to partially depend on.
- This is why 2NF violations typically show up in junction/bridge tables built around a composite key, like Enroll(StudentID, CourseID, ...).
- Exam tip: look for the phrase "composite primary key" in the scenario — a single-column key almost always means the answer to "is this in 2NF?" is yes, assuming 1NF already holds.

Q2 Demo: 2NF removes partial dependency on the primary key

```
mysql> SELECT * FROM Enroll_BadDesign; -- StudentName depends only on StudentID (partial dependency)
+-----+-----+-----+-----+
| StudentID | CourseID | StudentName | Grade |
+-----+-----+-----+-----+
| 1         | 101      | Arun        | A     |
| 1         | 102      | Arun        | B     |
+-----+-----+-----+-----+
2 rows in set

mysql> SELECT * FROM Student; -- split out: no partial dependency now
+-----+-----+
| StudentID | StudentName |
+-----+-----+
| 1         | Arun        |
+-----+-----+
1 row in set
```

Q3. Which normal form deals with multi-valued dependencies?

Bloom's Level: BL2 – Understand | Marks: 5

- (a) 2NF
- (b) 3NF
- (c) BCNF

✓ (d) 4NF — CORRECT ANSWER

Explanation:

Fourth Normal Form (4NF) specifically targets multi-valued dependencies (MVDs) — situations where one key is associated with multiple independent sets of values (e.g., an employee's skills and the languages they speak), causing combinatorial redundancy that 2NF/3NF/BCNF do not address.

Additional Justification:

- The notation $A \twoheadrightarrow B$ ("A multi-determines B") says the set of B-values for a fixed A is the same no matter which value of some other attribute C it's paired with — that independence is what forces the row count to multiply.
- A relation can be in BCNF and still fail 4NF, because BCNF's rules only look at single-valued functional dependencies, not MVDs — that's why 4NF had to be defined as a separate level.
- Exam tip: two unrelated multi-valued facts about the same entity (skills and languages, hobbies and certifications) is the classic signature of a 4NF/MVD question.

```
Q3 Demo
mysql> SELECT * FROM Emp_MVD; -- Skill and Language are independent MVDs -> redundancy, violates 4NF
+-----+-----+-----+
| EmpID | Skill | Language |
+-----+-----+-----+
| 1     | SQL   | English  |
| 1     | SQL   | Tamil    |
| 1     | Python | English  |
| 1     | Python | Tamil    |
+-----+-----+-----+
4 rows in set

mysql> SELECT * FROM Emp_Skill; -- decomposed to 4NF, no redundant combinations
+-----+-----+
| EmpID | Skill |
+-----+-----+
| 1     | SQL   |
| 1     | Python |
+-----+-----+
2 rows in set
```

Q4. BCNF is a stricter version of:

Bloom's Level: BL2 – Understand | Marks: 5

- (a) 1NF
- (b) 2NF

✓ (c) 3NF — CORRECT ANSWER

- (d) 4NF

Explanation:

Boyce-Codd Normal Form (BCNF) tightens 3NF: in 3NF a non-prime attribute may still be non-transitively dependent through a determinant that is not a candidate key, but BCNF requires that EVERY determinant (left-hand side of a functional dependency) be a candidate key. Every relation in BCNF is automatically in 3NF, but not vice versa.

Additional Justification:

- The formal BCNF test: for every non-trivial FD $X \rightarrow Y$ in the relation, X must be a superkey. One failing FD is enough to disqualify the whole relation from BCNF.
- Decomposing into BCNF always removes redundancy, but can sometimes lose dependency preservation — a reason designers occasionally settle for 3NF instead when preserving every constraint matters more.

- Exam tip: a determinant that is part of the key but not the whole key (like Course, part of StudentID+Course) is the textbook 3NF-but-not-BCNF pattern.

```
Q4 Demo
mysql> SELECT * FROM Course_3NF_not_BCNF;  -- Course->Instructor holds, Course is not a candidate key: violates BCNF
+-----+-----+-----+
| StudentID | Course | Instructor |
+-----+-----+-----+
| 1         | DBMS   | Dr.Kumar   |
| 2         | DBMS   | Dr.Kumar   |
+-----+-----+-----+
2 rows in set

mysql> SELECT * FROM Course_Instructor;  -- every determinant is now a candidate key: satisfies BCNF
+-----+-----+
| Course | Instructor |
+-----+-----+
| DBMS   | Dr.Kumar   |
+-----+-----+
1 row in set
```

Q5. Functional dependency $X \rightarrow Y$ means:

Bloom's Level: BL2 – Understand | Marks: 5

(a) Y determines X

✓ **(b) X determines Y — CORRECT ANSWER**

(c) X and Y are unrelated

(d) Both determine each other

Explanation:

$X \rightarrow Y$ (read 'X determines Y') means that for every value of X there is exactly one associated value of Y — knowing X tells you Y uniquely. It says nothing about Y determining X; that would be the separate dependency $Y \rightarrow X$.

Additional Justification:

- $X \rightarrow Y$ behaves like a mathematical function from X-values to Y-values: each X maps to exactly one Y, but the reverse mapping isn't guaranteed, since different X values can share the same Y (RollNo 1 and 3 both map to "Arun" below).
- X being a determinant for Y doesn't automatically make X a candidate key of the whole relation — that requires X to determine every other attribute, not just this one.
- Exam tip: to check if $A \rightarrow B$ holds in a data table, group rows by A and confirm every row in each group shares the same B value.

```
Q5 Demo
mysql> SELECT RollNo, Name FROM FD_Demo ORDER BY RollNo;  -- RollNo -> Name: each RollNo determines exactly one Name
+-----+-----+
| RollNo | Name |
+-----+-----+
| 1      | Arun |
| 2      | Bala |
| 3      | Arun |
+-----+-----+
3 rows in set
```

Q6. Which of the following anomaly is NOT associated with unnormalized relations?

Bloom's Level: BL2 – Understand | Marks: 5

(a) Insertion anomaly

(b) Deletion anomaly

(c) Update anomaly

✓ **(d) Retrieval anomaly — CORRECT ANSWER**

Explanation:

Unnormalized (redundant) tables classically suffer from insertion, deletion, and update anomalies — problems caused by storing the same fact in multiple rows. 'Retrieval anomaly' is not a recognized term in normalization theory; retrieving (SELECTing) data is unaffected by normalization the way modifying it is.

Additional Justification:

- Normalization protects write-side consistency, not read correctness — a redundant table can still be read correctly, just with duplicate copies of the same fact.
- Normalized schemas can even make reads slower, since a full picture now needs a JOIN across decomposed tables — a performance cost, never called a "retrieval anomaly."
- Exam tip: the three real anomalies map onto INSERT, DELETE, UPDATE. A fourth option naming a different operation (SELECT/retrieval) is a fabricated distractor.

```
Q6 Demo
mysql> SELECT * FROM Unnormalized;  -- before delete: 'IT' dept info stored redundantly on every employee row
+-----+-----+-----+
| EmployeeID | Dept | Salary |
+-----+-----+-----+
| 1          | IT   | 50000  |
| 2          | IT   | 60000  |
| 3          | IT   | 70000  |
| 4          | IT   | 80000  |
| 5          | IT   | 90000  |
| 6          | IT   | 100000 |
| 7          | IT   | 110000 |
| 8          | IT   | 120000 |
| 9          | IT   | 130000 |
| 10         | IT   | 140000 |
| 11         | IT   | 150000 |
| 12         | IT   | 160000 |
| 13         | IT   | 170000 |
| 14         | IT   | 180000 |
| 15         | IT   | 190000 |
| 16         | IT   | 200000 |
| 17         | IT   | 210000 |
| 18         | IT   | 220000 |
| 19         | IT   | 230000 |
| 20         | IT   | 240000 |
| 21         | IT   | 250000 |
| 22         | IT   | 260000 |
| 23         | IT   | 270000 |
| 24         | IT   | 280000 |
| 25         | IT   | 290000 |
| 26         | IT   | 300000 |
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| 30         | IT   | 340000 |
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| 33         | IT   | 370000 |
| 34         | IT   | 380000 |
| 35         | IT   | 390000 |
| 36         | IT   | 400000 |
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| 38         | IT   | 420000 |
| 39         | IT   | 430000 |
| 40         | IT   | 440000 |
| 41         | IT   | 450000 |
| 42         | IT   | 460000 |
| 43         | IT   | 470000 |
| 44         | IT   | 480000 |
| 45         | IT   | 490000 |
| 46         | IT   | 500000 |
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| 49         | IT   | 530000 |
| 50         | IT   | 540000 |
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| 52         | IT   | 560000 |
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| 72         | IT   | 760000 |
| 73         | IT   | 770000 |
| 74         | IT   | 780000 |
| 75         | IT   | 790000 |
| 76         | IT   | 800000 |
| 77         | IT   | 810000 |
| 78         | IT   | 820000 |
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| 84         | IT   | 880000 |
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| 87         | IT   | 910000 |
| 88         | IT   | 920000 |
| 89         | IT   | 930000 |
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| 91         | IT   | 950000 |
| 92         | IT   | 960000 |
| 93         | IT   | 970000 |
| 94         | IT   | 980000 |
| 95         | IT   | 990000 |
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| 98         | IT   | 1020000 |
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| 406        | IT   | 4100000 |
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| 442        | IT   | 4460000 |
| 443        | IT   | 4470000 |
| 444        | IT   | 4480000 |
| 445        | IT   | 4490000 |
| 446        | IT   | 4500000 |
| 447        | IT   | 4510000 |
| 448        | IT   | 4520000 |
| 449        | IT   | 4530000 |
| 450        | IT   | 4540000 |
| 451        | IT   | 4550000 |
| 452        | IT   | 4560000 |
| 453        | IT   | 4570000 |
| 454        | IT   | 4580000 |
| 455        | IT   | 4590000 |
| 456        | IT   | 4600000 |
| 457        | IT   | 4610000 |
| 458        | IT   | 4620000 |
| 459        | IT   | 4630000 |
| 460        | IT   | 4640000 |
| 461        | IT   | 4650000 |
| 462        | IT   | 4660000 |
| 463        | IT   | 4670000 |
| 464        | IT   | 4680000 |
| 465        | IT   | 4690000 |
| 466        | IT   | 4700000 |
| 467        | IT   | 4710000 |
| 468        | IT   | 4720000 |
| 469        | IT   | 4730000 |
| 470        | IT   | 4740000 |
| 471        | IT   | 4750000 |
| 472        | IT   | 4760000 |
| 473        | IT   | 4770000 |
| 474        | IT   | 4780000 |
| 475        | IT   | 4790000 |
| 476        | IT   | 4800000 |
| 477        | IT   | 4810000 |
| 478        | IT   | 4820000 |
| 479        | IT   | 4830000 |
| 480        | IT   | 4840000 |
| 481        | IT   | 4850000 |
| 482        | IT   | 4860000 |
| 483        | IT   | 4870000 |
| 484        | IT   | 4880000 |
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| 486        | IT   | 4900000 |
| 487        | IT   | 4910000 |
| 488        | IT   | 4920000 |
| 489        | IT   | 4930000 |
| 490        | IT   | 4940000 |
| 491        | IT   | 4950000 |
| 492        | IT   | 4960000 |
| 493        | IT   | 4970000 |
| 494        | IT   | 4980000 |
| 495        | IT   | 4990000 |
| 496        | IT   | 5000000 |
| 497        | IT   | 5010000 |
| 498        | IT   | 5020000 |
| 499        | IT   | 5030000 |
| 500        | IT   | 5040000 |
| 501        | IT   | 5050000 |
| 502        | IT   | 5060000 |
| 503        | IT   | 5070000 |
| 504        | IT   | 5080000 |
| 505        | IT   | 5090000 |
| 506        | IT   | 5100000 |
| 507        | IT   | 5110000 |
| 508        | IT   | 5120000 |
| 509        | IT   | 5130000 |
| 510        | IT   | 5140000 |
| 511        | IT   | 5150000 |
| 512        | IT   | 5160000 |
| 513        | IT   | 5170000 |
| 514        | IT   | 5180000 |
| 515        | IT   | 5190000 |
| 516        | IT   | 5200000 |
| 517        | IT   | 5210000 |
| 518        | IT   | 5220000 |
| 519        | IT   | 5230000 |
| 520        | IT   | 5240000 |
| 521        | IT   | 5250000 |
| 522        | IT   | 5260000 |
| 523        | IT   | 5270000 |
| 524        | IT   | 5280000 |
| 525        | IT   | 5290000 |
| 526        | IT   | 5300000 |
| 527        | IT   | 5310000 |
| 528        | IT   | 5320000 |
| 529        | IT   | 5330000 |
| 530        | IT   | 5340000 |
| 531        | IT   | 5350000 |
| 532        | IT   | 5360000 |
| 533        | IT   | 5370000 |
| 534        | IT   | 5380000 |
| 535        | IT   | 5390000 |
| 536        | IT   | 5400000 |
| 537        | IT   | 5410000 |
| 538        | IT   | 5420000 |
| 539        | IT   | 5430000 |
| 540        | IT   | 5440000 |
| 541        | IT   | 5450000 |
| 542        | IT   | 5460000 |
| 543        | IT   | 5470000 |
| 544        | IT   | 5480000 |
| 545        | IT   | 5490000 |
| 546        | IT   | 5500000 |
|
```

```
| EmpID | EmpName | DeptName |
+-----+-----+
| 1     | Arun    | IT       |
| 2     | Bala    | IT       |
+-----+-----+
2 rows in set
```

```
mysql> SELECT * FROM Unnormalized; -- after deleting Arun (only remaining row shown): if Bala didn't exist, 'IT' dept fact would vanish
+-----+-----+
| EmpID | EmpName | DeptName |
+-----+-----+
| 2     | Bala    | IT       |
+-----+-----+
1 row in set
```

Q7. A superkey is:

Bloom's Level: BL2 – Understand | Marks: 5

- (a) A minimal set of attributes that uniquely identifies tuples
- ✓ **(b) Any set of attributes that uniquely identifies tuples — CORRECT ANSWER**
- (c) An attribute that depends on a non-prime attribute
- (d) A foreign key reference

Explanation:

A superkey is ANY combination of attributes whose values uniquely identify a row — it need not be minimal. Once you strip a superkey down so that no attribute can be removed without losing uniqueness, it becomes a candidate key. Option (a) actually describes a candidate key, not a superkey.

Additional Justification:

- Full hierarchy: every candidate key is a superkey, but not every superkey is a candidate key; the primary key is whichever candidate key the designer picks; a foreign key is a separate concept referencing another table's key.
- Mental test: remove attributes from a superkey one at a time — if uniqueness survives, keep removing. It stops being reducible right at the candidate key.
- Exam tip: the word "minimal" in an option is what marks it as describing a candidate key rather than a superkey.

```
Q7 Demo
mysql> SELECT DISTINCT RollNo, Email FROM SuperKeyDemo; -- {RollNo,Email} uniquely identifies rows: it's a superkey
+-----+-----+
| RollNo | Email   |
+-----+-----+
| 1      | a@x.com |
| 2      | b@x.com |
+-----+-----+
2 rows in set
```

Q8. If R(A,B,C) has FD: A->B and B->C, then C is:

Bloom's Level: BL2 – Understand | Marks: 5

- (a) Partially dependent on A
- (b) Fully dependent on A
- ✓ **(c) Transitively dependent on A — CORRECT ANSWER**
- (d) Independently determined

Explanation:

Because A->B and B->C, we get A->C only indirectly, through the intermediate attribute B. This chain (A determines C only via B, where B is not part of a key) is the textbook definition of a transitive dependency, and it is exactly what 3NF removes.

Additional Justification:

- Armstrong's transitivity axiom formally states: if $A \rightarrow B$ and $B \rightarrow C$ both hold, $A \rightarrow C$ is logically implied, even if never written down explicitly.
- This only counts as a 3NF-violating transitive dependency when B is non-prime (not part of a key) — if B were part of the key, the chain wouldn't violate 3NF.
- Exam tip: spotting two FDs sharing a middle attribute ($A \rightarrow B$, $B \rightarrow C$) is the standard pattern for identifying a transitive dependency.

```
Q8 Demo
mysql> SELECT * FROM Transitive_Demo; -- A->B and B->C, so A->C holds only via B: C is transitively dependent on A
+-----+-----+
| A | B | C |
+-----+-----+
| 1 | B1 | C1 |
| 2 | B1 | C1 |
+-----+-----+
2 rows in set
```

Q9. 5NF is also known as:

Bloom's Level: BL2 – Understand | Marks: 5

- (a) Domain-Key Normal Form
- (b) Boyce-Codd Normal Form
- ✓ (c) Project-Join Normal Form — CORRECT ANSWER
- (d) Element Normal Form

Explanation:

Fifth Normal Form (5NF) is also called Project-Join Normal Form (PJNF) because it is defined in terms of join dependencies: a relation is in 5NF when every join dependency in it is implied by its candidate keys, i.e., it cannot be decomposed further without losing information on rejoin.

Additional Justification:

- Normal-form ladder: 1NF → 2NF → 3NF → BCNF → 4NF → 5NF/PJNF → DKNF, each tightening the previous level's constraint.
- A three-way join dependency (like Agent/Company/Product below) needs three tables rejoined together to reconstruct losslessly — different from a two-way MVD, which only needs two tables. That's why 5NF is its own level, separate from 4NF.
- Exam tip: three or more interrelated entities that must all rejoin to avoid spurious rows signals 5NF, not 4NF.

```
Q9 Demo
mysql> SELECT * FROM Supply; -- join-dependency exists among 3 attributes: decompose losslessly to reach 5NF
+-----+-----+-----+
| Agent | Company | Product |
+-----+-----+-----+
| AgentA | CompanyX | Product1 |
| AgentA | CompanyY | Product1 |
+-----+-----+-----+
2 rows in set
```

Q10. Lossless-join decomposition ensures:

Bloom's Level: BL2 – Understand | Marks: 5

- ✓ (a) No data is lost during decomposition — CORRECT ANSWER
- (b) All FDs are preserved
- (c) The relation is in BCNF
- (d) No null values exist

Explanation:

A lossless-join (non-additive-join) decomposition guarantees that when the decomposed relations are joined back together (natural join), you get back exactly the original relation — no spurious rows and no lost rows/data. It does not by itself guarantee dependency preservation or any particular normal form.

Additional Justification:

- Formal two-relation test: the decomposition is lossless iff the common attributes of R1 and R2 form a superkey of at least one of them — $(R1 \cap R2) \rightarrow R1$ or $(R1 \cap R2) \rightarrow R2$.
- "Spurious rows" are the real risk: if the common attribute weren't a key of either side, rejoining could fabricate combinations that never existed in the source data — arguably worse than losing rows, since a spurious row looks valid.
- Exam tip: don't conflate lossless-join with dependency preservation — they're checked by separate algorithms, and a decomposition can achieve one without the other.

```
Q10 Demo
mysql> SELECT * FROM Original;
+-----+-----+-----+
| StudentID | Course | Instructor |
+-----+-----+-----+
| 1          | DBMS   | Dr.Kumar   |
| 2          | DBMS   | Dr.Kumar   |
+-----+-----+-----+
2 rows in set

mysql> SELECT R1.StudentID, R1.Course, R2.Instructor FROM R1 JOIN R2 ON R1.Course=R2.Course; -- rejoin reproduces Original
+-----+-----+-----+
| StudentID | Course | Instructor |
+-----+-----+-----+
| 1          | DBMS   | Dr.Kumar   |
| 2          | DBMS   | Dr.Kumar   |
+-----+-----+-----+
2 rows in set
```

Answer Key Summary

Q.No	Correct Answer	Topic
Q1	(c) All attributes must be atomic	1NF – atomicity

Q2	(b) Has no partial dependencies on primary key	2NF – partial dependency
Q3	(d) 4NF	4NF – multi-valued dependency
Q4	(c) 3NF	BCNF vs 3NF
Q5	(b) X determines Y	Functional dependency notation
Q6	(d) Retrieval anomaly	Anomalies in unnormalized relations
Q7	(b) Any set of attributes that uniquely identifies tuples	Superkey definition
Q8	(c) Transitively dependent on A	Transitive dependency
Q9	(c) Project-Join Normal Form	5NF / PJNF
Q10	(a) No data is lost during decomposition	Lossless-join decomposition

Code of Conduct

I K. Anil Kumar Reddy (Name / Reg No: 192525239) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
 I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____